The genus-2 free energy of W_3 and the gravitational coproduct

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Remark 0.1 (Relationship to companion papers)*.* The W_3 composite field identity $\Lambda_0|h\rangle = h^2 - 9h/5$ and the complete rank-2 holographic datum are developed in the companion paper *The first rank-2 holographic datum:* $\mathcal{H}(W_3)$ (`w3_holographic_datum.tex`).

Remark 0.2 (Wave-14 reconstitution anchor)*.* This note records the graph-by-graph computation that produces the genus-2 cross-channel correction $\delta F_2(W_3) = (c + 204)/(16c)$ and the gravitational-coproduct primitivity for Vir. Two canonical compute modules carry the independent verification paths: `w3_genus2.py` (six-graph tabulation at seven test central charges) and `ds_hpl_primitivity.py` (ghost-number obstruction check). The exceptional-Lie dimensions used elsewhere in the monograph are not re-hardcoded here; when an exceptional dimension appears it is imported from `compute/lib/bc_exceptional_categorical_zeta_engine.py` at `FUNDAMENTAL_DIMS['E8']`. Cross-reference to Vol I theorem status: thm:multi-weight-genus-expansion gives $\delta F_2(W_3) = (c+204)/(16c)$ explicit (proved, not conjectural); thm:grav-primitivity-standalone is the ghost-number-obstruction argument repeated here in three steps. The conjectural extension to general principal $\mathcal{W}(\mathfrak{g})$ is Conjecture 2.3.

1 The genus-2 free energy of W_3

The preceding sections treated the single-channel regime: one strong generator, one primary line, one modular characteristic κ. We now compute the first multi-channel invariant. The W_3 algebra has two strong generators (the stress tensor T of weight 2 and the spin-3 current W of weight 3), and $F_2(W_3)$ receives contributions from both channels and their interactions.

1.1 The Frobenius algebra of the bar-complex CohFT

The bar-complex CohFT of W_3 has a two-dimensional Frobenius algebra with basis $\{e_T, e_W\}$ and diagonal metric from the per-channel modular characteristics:

$$\eta_{TT} = \kappa_T = \frac{c}{2}, \qquad \eta_{WW} = \kappa_W = \frac{c}{3}, \qquad \eta_{TW} = 0. \tag{1}$$

The total modular characteristic is $\kappa(W_3) = \kappa_T + \kappa_W = 5c/6 = c \cdot (H_3 - 1)$, where $H_3 = 1 + 1/2 + 1/3 = 11/6$ is the third harmonic number. The inverse metric gives the propagators:

$$\eta^{TT} = \frac{2}{c}, \qquad \eta^{WW} = \frac{3}{c}. \tag{2}$$

Both propagators have weight 1: the bar-complex propagator is $d\log E(z, w)$, which has weight 1 regardless of the conformal weight of the field being sewed.

The OPE determines the three-point structure constants. The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ kills all three-point functions with an odd number of W-insertions:

$$C_{TTT} = C_{TWW} = C_{WWT} = c, \qquad C_{TTW} = C_{TWT} = C_{WTT} = C_{WWW} = 0. \tag{3}$$

The derivations are: $T_{(1)}T = 2T$ gives $C^T_{TT} = 2$, whence $C_{TTT} = \eta_{TT} \cdot 2 = c$; $T_{(1)}W = 3W$ gives $C^W_{TW} = 3$, whence $C_{TWW} = \eta_{WW} \cdot 3 = c$; $W_{(3)}W = 2T$ gives $C^T_{WW} = 2$, whence $C_{WWT} = \eta_{TT} \cdot 2 = c$.

The genus-0 four-point vertex is universal: for any channel pair $(i, j) \in \{T, W\}^2$,

$$V_{0,4}(i, i \mid j, j) \;=\; \sum_m \eta^{mm} C_{iim} C_{jjm} \;=\; \eta^{TT} \cdot c \cdot c \;=\; 2c. \tag{4}$$

Only the T-channel contributes to the sum: $C_{iiT} = c$ for $i \in \{T, W\}$, while $C_{iiW} = 0$ for both $i = T$ (since $C_{TTW} = 0$) and $i = W$ (since $C_{WWW} = 0$). The W-channel is invisible at genus-0 valence 4.

1.2 The seven stable graphs at genus 2

The moduli space $\overline{\mathcal{M}}_{2,0}$ has a stratification by seven stable graphs $\Gamma_0, \ldots, \Gamma_6$:

	Name	**Vertices**	**Edges**	$\lvert\mathrm{Aut}\rvert$	h^1	**Type**
Γ_0	smooth	(g=2, val=0)	0	1	0	compact stratum
Γ_1	figure-eight	(g=1, val=2)	1 self-loop	2	1	one-loop
Γ_2	banana	(g=0, val=4)	2 self-loops	8	2	two-loop sunset
Γ_3	dumbbell	(g=1, val=1) + (g=1, val=1)	1 bridge	2	0	separating
Γ_4	theta	(g=0, val=3) + (g=0, val=3)	3 bridges	12	2	non-separating
Γ_5	tadpole	(g=0, val=3) + (g=1, val=1)	1 self-loop + 1 bridge	2	1	mixed
Γ_6	chain	(g=0, val=3) + (g=0, val=3)	2 self-loops + 1 bridge	8	2	non-separating

Every vertex satisfies the stability condition $2g_v + \mathrm{val}_v \geq 3$, and the genus formula $h^1(\Gamma) + \sum_v g_v = 2$ holds in each case. Note that the smooth graph Γ_0 has no edges and contributes the integrated class on $\mathcal{M}_{2,0}$ itself; the remaining six graphs are the boundary strata.

1.3 Multi-channel Feynman rules

For each graph Γ with $|E(\Gamma)|$ edges, a *channel assignment* is a function $\sigma\colon E(\Gamma) \to \{T, W\}$. The Feynman rules of the bar-complex CohFT assign to each pair (Γ, σ) an amplitude

$$\mathcal{A}(\Gamma, \sigma) \;=\; \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e), \sigma(e)} \prod_{v \in V(\Gamma)} V_v\big(g_v;\, \sigma|_{\text{half-edges}(v)}\big), \tag{5}$$

where the vertex factors are:

- *Genus-0, trivalent*: $V_v = C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.
- *Genus-0, four-valent* (the banana vertex): $V_v = \sum_m \eta^{mm} C_{\sigma(e_1)\sigma(e_2)m}\, C_{m\,\sigma(e_3)\sigma(e_4)}$.

- *Genus-1, valence n*: $V_v = \kappa_{\sigma(e)}/24$ (per-channel genus-1 universality, proved unconditionally).

Each half-edge at a self-loop contributes a pair of half-edges to the same vertex, both carrying channel $\sigma(e)$.

1.4 Per-graph amplitudes

We compute each boundary graph amplitude, decomposed into *diagonal* (all edges in one channel) and *mixed* (edges in different channels) contributions.

Γ_1: figure-eight

One edge, two channel assignments. The vertex is genus-1 with valence 2, so the vertex factor for channel i is $\kappa_i/24$:

$$\mathcal{A}(\Gamma_1, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \tfrac{\kappa_T}{24} = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c}{48} = \tfrac{1}{48}, \tag{6}$$

$$\mathcal{A}(\Gamma_1, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \tfrac{\kappa_W}{24} = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c}{72} = \tfrac{1}{48}. \tag{7}$$

Both channels contribute equally: the propagator $1/\kappa_i$ cancels the genus-1 vertex factor $\kappa_i/24$, leaving the universal value $1/48$ independent of c. The total is $1/24 = \lambda_1^{\mathrm{FP}}$. No mixed channel assignments exist (single edge).

Γ_2: banana

Two edges, four channel assignments, $|\mathrm{Aut}| = 8$. The vertex is genus-0 with valence 4, and the four-point vertex factor is $V_{0,4}(i, i \mid j, j) = 2c$ universally (4).

$$\mathcal{A}(\Gamma_2, TT) = \tfrac{1}{8}\,(\eta^{TT})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{4}{c^2} \cdot 2c = \frac{1}{c}, \tag{8}$$

$$\mathcal{A}(\Gamma_2, WW) = \tfrac{1}{8}\,(\eta^{WW})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{9}{c^2} \cdot 2c = \frac{9}{4c}, \tag{9}$$

$$\mathcal{A}(\Gamma_2, TW{+}WT) = \tfrac{1}{8} \cdot 2 \cdot \eta^{TT}\,\eta^{WW} \cdot 2c = \tfrac{1}{8} \cdot 2 \cdot \tfrac{6}{c^2} \cdot 2c = \frac{3}{c}. \tag{10}$$

The cross-channel contribution is $\delta_{\mathrm{banana}} = 3/c$.

Γ_3: dumbbell

One bridge edge, two channel assignments, $|\mathrm{Aut}| = 2$. Each vertex has genus 1 and valence 1, with vertex factor $\kappa_i/24$:

$$\mathcal{A}(\Gamma_3, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \left(\tfrac{\kappa_T}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c^2}{2304} = \frac{c}{2304}, \tag{11}$$

$$\mathcal{A}(\Gamma_3, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \left(\tfrac{\kappa_W}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c^2}{5184} = \frac{c}{3456}. \tag{12}$$

The total is $c/2304 + c/3456 = (3c + 2c)/6912 = 5c/6912 = \kappa(W_3)/1152$. No mixed channel assignments exist (single edge).

Γ_4: theta

Three bridge edges, eight channel assignments, $|\mathrm{Aut}| = 12$. Each vertex has genus 0 and valence 3, with vertex factor $C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.

- *All-T*: both vertices get $C_{TTT} = c$. Amplitude $= \frac{1}{12}(\eta^{TT})^3 \cdot c^2 = \frac{1}{12} \cdot \frac{8}{c^3} \cdot c^2 = \frac{2}{3c}$.
- *All-W*: both vertices get $C_{WWW} = 0$. Amplitude $= 0$.
- *Type* (T, T, W) (three such assignments): each vertex receives one W half-edge, giving $C_{TTW} = 0$. All three vanish.
- *Type* (T, W, W) (three such assignments): each vertex receives half-edges (T, W, W), giving $C_{TWW} = c$. The propagator product is $\eta^{TT}(\eta^{WW})^2 = (2/c)(9/c^2) = 18/c^3$. Each assignment contributes $18c^2/c^3 = 18/c$; with $|\mathrm{Aut}| = 12$, the total mixed amplitude is

$$\delta_{\mathrm{theta}} = \frac{3 \cdot 18}{12c} = \frac{9}{2c}. \tag{13}$$

Γ_5: tadpole

One self-loop at vertex v_0 (genus 0, valence 3) and one bridge to vertex v_1 (genus 1, valence 1). Two edges, four channel assignments, $|\mathrm{Aut}| = 2$.

- (T, T): self-loop T, bridge T. Vertex v_0 gets half-edges $(T, T, T) \to C_{TTT} = c$. Vertex v_1 gets half-edge $T \to \kappa_T/24 = c/48$. Propagators: $(\eta^{TT})^2 = 4/c^2$. With $|\mathrm{Aut}|$: $\frac{1}{2} \cdot \frac{4}{c^2} \cdot c \cdot \frac{c}{48} = \frac{1}{24}$.
- (W, W): self-loop W, bridge W. Vertex v_0 gets $(W, W, W) \to C_{WWW} = 0$. Vanishes.
- (T, W): self-loop T, bridge W. Vertex v_0 gets $(T, T, W) \to C_{TTW} = 0$. Vanishes.
- (W, T): self-loop W, bridge T. Vertex v_0 gets $(W, W, T) \to C_{WWT} = c$. Vertex v_1 gets $T \to \kappa_T/24 = c/48$. Propagators: $\eta^{WW} \cdot \eta^{TT} = (3/c)(2/c) = 6/c^2$. With $|\mathrm{Aut}|$:

$$\delta_{\mathrm{tadpole}} = \tfrac{1}{2} \cdot \tfrac{6}{c^2} \cdot c \cdot \tfrac{c}{48} = \tfrac{6c^2}{96c^2} = \frac{1}{16}. \tag{14}$$

The tadpole mixed amplitude $1/16$ is *independent of* c: the c^2 in the numerator (from $C_{WWT} \cdot \kappa_T/24$) cancels the c^2 in the denominator (from $\eta^{WW} \cdot \eta^{TT}$).

Γ_6: chain (barbell)

Two genus-0 trivalent vertices, each carrying one self-loop, connected by a bridge. Three edges (self-loop s_1, self-loop s_2, bridge b), eight channel assignments, $|\mathrm{Aut}| = 8$. Each vertex v_j receives half-edges $(\sigma(s_j), \sigma(s_j), \sigma(b))$ with vertex factor $C_{\sigma(s_j)\,\sigma(s_j)\,\sigma(b)}$.

Only $\sigma(b) = T$ gives nonzero vertex factors (since $C_{TTW} = C_{WWW} = 0$). Among the four remaining assignments $(\sigma(s_1), \sigma(s_2)) \in \{T, W\}^2$ with $\sigma(b) = T$:

- (T, T, T): diagonal. Vertex factors $C_{TTT}^2 = c^2$, propagators $(\eta^{TT})^3 = 8/c^3$. With $|\mathrm{Aut}|$: $(1/8)(8/c^3)(c^2) = 1/c$.

- (T,W,T): mixed. $C_{TTT} \cdot C_{WWT} = c^2$, $\eta^{TT}\,\eta^{WW}\,\eta^{TT} = 12/c^3$. With $|\mathrm{Aut}|$: $(1/8)(12/c^3)(c^2) = 3/(2c)$.
- (W,T,T): mixed. By the $s_1 \leftrightarrow s_2$ symmetry: $3/(2c)$.
- (W,W,T): mixed. $C_{WWT}^2 = c^2$, $(\eta^{WW})^2\,\eta^{TT} = 18/c^3$. With $|\mathrm{Aut}|$: $(1/8)(18/c^3)(c^2) = 9/(4c)$.

The total mixed amplitude is

$$\delta_{\mathrm{barbell}} = \frac{3}{2c} + \frac{3}{2c} + \frac{9}{4c} = \frac{6+6+9}{4c} = \frac{21}{4c}. \tag{15}$$

1.5 The cross-channel correction

Collecting the mixed-channel amplitudes from all boundary graphs:

$$\delta F_2 \;=\; \underbrace{\frac{3}{c}}_{\text{banana}} \;+\; \underbrace{\frac{9}{2c}}_{\text{theta}} \;+\; \underbrace{\frac{1}{16}}_{\text{tadpole}} \;+\; \underbrace{\frac{21}{4c}}_{\text{barbell}} \;=\; \frac{51}{4c} + \frac{1}{16} \;=\; \frac{c+204}{16c}. \tag{16}$$

The figure-eight and dumbbell graphs contribute zero mixed amplitude (single edges admit only diagonal assignments). The correction decomposes into two structurally distinct pieces:

- A *c-dependent* piece $51/(4c)$, arising from genus-0 vertex factors on the banana ($3/c$), theta ($9/(2c)$), and barbell ($21/(4c)$) graphs. This is R-matrix independent: genus-0 vertices receive no corrections from the CohFT R-matrix.
- A *c-independent* piece $1/16$, arising from the tadpole graph, which mixes a genus-0 vertex factor with a genus-1 vertex factor $\kappa_i/24$.

Remark 1.1 (Comparison with the scalar approximation). The per-channel (diagonal) graph sum, computed by applying Theorem D to each channel independently, gives

$$F_2^{\mathrm{scal}}(W_3) = \kappa \cdot \lambda_2^{\mathrm{FP}} = \frac{5c}{6} \cdot \frac{7}{5760} = \frac{7c}{6912}. \tag{17}$$

The ratio of the naive cross-channel correction to the scalar approximation is

$$\frac{\delta F_2}{F_2^{\mathrm{scal}}} = \frac{432(c+204)}{7c^2}, \tag{18}$$

which diverges as $c \to 0$ and decays as c^{-1} at large c. At $c = 4$: the ratio is approximately 802, and the cross-channel correction *dominates* the scalar term. At $c = 50$: the ratio is approximately 6.3. At large c, the tadpole's c-independent contribution $1/16$ dominates, and the correction becomes a subleading $O(1/c)$ effect relative to the $O(c)$ scalar term.

Remark 1.2 (Status of the cross-channel correction). The computation above uses the naive genus-1 vertex factors $\kappa_i/24$ without R-matrix corrections. The multi-generator universality problem (**op:multi-generator-universality**) is resolved negatively: the CohFT R-matrix does *not* produce compensating terms that cancel δF_2. The cross-channel correction $\delta F_2 = (c{+}204)/(16c)$ is a genuine additional term, and the correct decomposition is $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$.

1.6 The complete amplitude table

We record the full per-graph data:

Graph	**Diagonal (T)**	**Diagonal (W)**	**Mixed**	**Total**
Γ_1 figure-eight	$1/48$	$1/48$	0	$1/24$
Γ_2 banana	$1/c$	$9/(4c)$	$3/c$	$25/(4c)$
Γ_3 dumbbell	$c/2304$	$c/3456$	0	$5c/6912$
Γ_4 theta	$2/(3c)$	0	$9/(2c)$	$31/(6c)$
Γ_5 tadpole	$1/24$	0	$1/16$	$5/48$
Γ_6 barbell	$1/c$	0	$21/(4c)$	$25/(4c)$

The table records the six boundary-graph amplitudes; the smooth graph Γ_0 has no edges and contributes an integrated class on $\mathcal{M}_{2,0}$ itself. The mixed column sums to $\delta F_2 = (c+204)/(16c)$.

Every entry in this table has been verified against the compute module `w3_genus2.py`, using exact rational arithmetic (`fractions.Fraction`), at the seven test central charges $c \in \{1, 2, 4, 13, 26, 50, 100\}$.

1.7 Planted-forest corrections

The planted-forest correction on each primary line is $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$.

T-line: $S_3^T = 2$ (the Virasoro cubic shadow), $\kappa_T = c/2$.

$$\delta_{\mathrm{pf}}^T = \frac{2(20 - c/2)}{48} = \frac{40 - c}{48}. \tag{19}$$

This vanishes at $c = 40$, is positive for $c < 40$, and negative for $c > 40$.

W-line: $S_3^W = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$, which forces all odd-degree shadows on the W-line to vanish. Hence $\delta_{\mathrm{pf}}^W = 0$.

The total planted-forest correction is $\delta_{\mathrm{pf}} = (40 - c)/48$, living entirely on the T-line.

1.8 Koszul duality constraints

Under the Koszul involution, W_3 at central charge c is dual to W_3 at central charge $c' = 100 - c$ (the Feigin–Frenkel involution for $\mathfrak{sl}_3$). The modular characteristics satisfy

$$\kappa(c) + \kappa(100 - c) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{250}{3}, \tag{20}$$

and the scalar free energies satisfy $F_2^{\mathrm{scal}}(c) + F_2^{\mathrm{scal}}(100 - c) = (250/3) \cdot \lambda_2^{\mathrm{FP}}$. The cross-channel correction at the dual pair is

$$\delta F_2(c) + \delta F_2(100 - c) = \frac{c + 204}{16c} + \frac{304 - c}{16(100 - c)} = -\frac{c^2 - 100c - 10200}{8c(100 - c)}.$$

This sum is not constant in c: the cross-channel correction does not respect Koszul additivity. If universality holds (δF_2 is cancelled by R-matrix corrections), this is automatic. If the naive correction persists, it produces a Koszul-asymmetric contribution to F_2: an observable departure from single-channel behaviour.

2 The gravitational coproduct is primitive

In gauge theory, gluons split: one gluon can exit a scattering region as two, and the dg-shifted Yangian coproduct encodes this splitting. In gravity, the graviton is composite, not fundamental. The stress tensor is the Sugawara quadratic

$$T = \frac{1}{k+2}\Big(\tfrac{1}{2}{:}J^0 J^0{:} + {:}J^+ J^-{:}\Big), \tag{21}$$

built from affine currents by normal ordering. This compositeness has a precise algebraic consequence: the transferred coproduct is trivial.

2.1 The DS-HPL transfer theorem

The Drinfeld–Sokolov reduction $\hat{\mathfrak{sl}}_2 \to \mathrm{Vir}$ is a BRST reduction with a strong deformation retract (i, p, h) from the BRST complex to the Virasoro cohomology. The homotopy h has ghost-number degree -1: it maps $V \mapsto b$ (ghost number $0 \to -1$) and $c \mapsto U/2$ (ghost number $1 \to 0$).

Theorem 2.1 (DS-HPL transfer)**.** *Let $Y^{\mathrm{dg}}(\widehat{\mathfrak{sl}}_2)$ denote the dg-shifted Yangian at level k, with A_∞ products $\{m_n^{\mathrm{aff}}\}$, coproduct Δ_z^{aff}, and r-matrix $r^{\mathrm{aff}}(z) = \Omega/((k{+}2)z)$. The HPL transfer through the BRST SDR produces:*

- (i) *Transferred A_∞ products $\{m_n^{\mathrm{Vir}}\}_{n\geq 2}$ on the Virasoro sector, with $m_n^{\mathrm{Vir}} \neq 0$ for all $n \geq 2$ (the entire infinite A_∞ tower of class* **M***).*
- (ii) *A transferred coproduct $\Delta_z^{\mathrm{Vir}} \colon H_{\mathrm{Vir}} \to H_{\mathrm{Vir}} \otimes H_{\mathrm{Vir}}$.*
- (iii) *A transferred r-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, with cubic leading pole reflecting the Sugawara nonlinearity.*

The affine r-matrix $r^{\mathrm{aff}}(z) = \Omega/((k{+}2)z)$ has a simple pole (the OPE has at most a double pole; the bar kernel $d\log(z-w)$ absorbs one power). The Virasoro OPE has a quartic pole (z^{-4} in $T(z)T(w)$); absorption produces the cubic pole z^{-3} of $r^{\mathrm{Vir}}(z)$. The general principle: r-matrix pole order = OPE pole order minus one.

Theorem 2.2 (Gravitational coproduct primitivity)**.** *The HPL-transferred coproduct is strictly primitive at all degrees:*

$$\Delta_{z,k}^{\mathrm{Vir}} = 0 \qquad \textit{for all } k \geq 2. \tag{22}$$

That is, $\Delta_z^{\mathrm{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. All nontrivial structure of the Virasoro dg-shifted Yangian resides in the $r(z)$-twisted target product, never in the coproduct.

Proof. The proof is a ghost-number obstruction, proceeding in three steps.

Step 1. The linear transferred coproduct $(p \otimes p)(\Delta_z(T))$ is already primitive. The affine coproduct $\Delta_z(T)$ expands as $\tau_z(T) \otimes 1 + 1 \otimes T + \frac{1}{k+2}\,\Omega(w{+}z,\, w)$, where $\Omega(w{+}z, w) = J^0(w{+}z) \otimes J^0(w) + J^+(w{+}z) \otimes J^-(w) + J^-(w{+}z) \otimes J^+(w)$. The projection p annihilates J^0 (not in the Virasoro sector) and J^+ decomposes as $V + 1$ with $p(V) = 0$. Every summand of Ω contains at least one factor from $\{J^0, J^-, J^+\}$ at conformal weight 1, where the Virasoro sector has no states ($L_{-1}|0\rangle = 0$). Each tensor factor of $(p \otimes p)(\Omega)$ has p killing at least one component, so $(p \otimes p)(\Omega) = 0$.

Step 2. At degree $k \geq 2$, every HPL tree has at least one internal edge labelled by the homotopy h. The ghost-number accounting is:

Operation	Ghost degree
h (homotopy)	-1
δ (BRST perturbation)	$+1$
μ (vertex algebra product)	0
Δ_z (coproduct)	0

The product $h\delta$ has ghost degree $-1+1=0$, but the tree topology at degree $k \geq 2$ always contains one *uncompensated* δ: a binary tree with k leaves has $k-1$ internal vertices (each carrying δ, ghost $+1$) and $k-2$ internal edges (each carrying h, ghost -1), giving total ghost number $(k-1)-(k-2)=+1$. The overall output before projection has ghost number $+1$.

Step 3. The W-algebra cohomology sits in ghost number 0. The projection p annihilates all ghost-number-($\neq 0$) elements. Since the output of each degree-k tree has ghost number $+1$, the projection $(p \otimes p)$ kills every summand.

The perturbation does not produce corrections. The BRST perturbation δ has ghost degree $+1$, so the product $h\delta$ preserves ghost number. But $\delta(T)=0$ (the stress tensor is BRST-closed), so $(h\delta)(i(T))=0$, and by induction $(h\delta)^n(i(T))=0$ for all $n \geq 1$. The corrected inclusion $i_\infty(T)=i(T)$ is unperturbed, and the same primitive coproduct results at every perturbative order. □

Conjecture 2.3 (General DS primitivity)**.** *For any simple Lie algebra $\mathfrak{g}$, the HPL-transferred coproduct of the principal $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g})$ is strictly primitive at all degrees. The ghost-number obstruction of the BRST complex for the principal Drinfeld–Sokolov reduction $V_k(\mathfrak{g}) \to \mathcal{W}_k(\mathfrak{g})$ forces $\Delta^{\mathcal{W}}_{z,n}=0$ for $n \geq 2$.*

2.2 Physical meaning

The coproduct Δ_z of a dg-shifted Yangian encodes the *splitting* of excitations. Its strictness or nontriviality is the algebraic fingerprint of the distinction between fundamental and composite fields.

Theory	**Boundary algebra**	$\boldsymbol{\Delta_z}$	**Shadow depth**	**Reason**
Chern–Simons	$\hat{\mathfrak{g}}_k$ (affine KM)	$\neq \varepsilon$	3 (class $\mathbf{L}$)	fundamental gluons split
3d gravity	$\mathrm{Vir}_c = W_k(\mathfrak{sl}_2)$	$= \varepsilon$	∞ (class $\mathbf{M}$)	ghost obstruction (DS)
Twisted holography	H_k (Heisenberg)	$= \varepsilon$	2 (class $\mathbf{G}$)	free field, $r = k/z$
M2 brane	DDCA	$\neq \varepsilon$	—	no DS, fundamental modes
M5 brane	$W_{1+\infty}(K)$	$= \varepsilon$	∞ (class $\mathbf{M}$)	ghost obstruction (DS)

The pattern: *principal Drinfeld–Sokolov reductions produce primitive coproducts.* The BRST ghost sector provides the grading obstruction that kills all higher-degree components. Theorem 2.2 proves this for $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$; the same argument applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$ (the ghost-number grading is universal), yielding Conjecture 2.3. Theories with fundamental excitations (Chern–Simons, deformed double current algebras) lack a ghost sector, and their coproducts are nontrivial.

The two exceptions to the DS pattern are illuminating. The Heisenberg algebra has a primitive coproduct for a different reason: it is a free field, its r-matrix $r^H(z)=k/z$ is scalar, and the bar complex is formal at degree 2 (shadow depth $r_{\max}=2$, class $\mathbf{G}$). No DS reduction and no ghost obstruction; primitivity follows from the absence of nonlinear OPE structure. The M2 brane boundary algebra (a deformed double current algebra) has a nontrivial coproduct because it is not obtained by DS reduction from any affine algebra; its excitations are fundamental modes of the M2 worldvolume theory.

2.3 The r-matrix carries all gravitational structure

With the coproduct primitive, the full gravitational content of the Virasoro dg-shifted Yangian is encoded in the r-matrix

$$r^{\mathrm{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{23}$$

The two terms encode distinct physical data:

- The *leading coefficient* $c/2 = \kappa(\mathrm{Vir}_c)$ is the modular characteristic. It determines the genus tower $F_g = (c/2)\int_{\overline{\mathcal{M}}_g} \lambda_g$ (uniform-weight) at all genera, the BTZ black hole entropy $S_{\mathrm{BTZ}} = 2\pi\sqrt{ch/6}$, and the Hawking–Page transition at $h = c/24$.
- The *subleading term* $2T/z$ encodes the state dependence of gravitational scattering. The braiding of Wilson lines (gravitational line operators) in the holomorphic-topological bulk is controlled by this term.
- The *pole structure* (cubic leading pole, simple subleading pole, no even poles) is the signature of a bosonic algebra extracted through the $d\log$ kernel: the OPE poles z^{-4}, z^{-2}, z^{-1} each lose one order under the $d\log$ extraction, producing r-matrix terms z^{-3}, z^{-1}, z^0. The z^0 term (from the ∂T pole at z^{-1}) is regular and does not contribute to the r-matrix, leaving only z^{-3} and z^{-1}.

The contrast with Chern–Simons is maximal. In Chern–Simons gauge theory, the affine r-matrix $r^{\mathrm{aff}}(z) = \Omega/((k+h^\vee)z)$ is a simple pole and the coproduct is nontrivial: gluons split and the splitting is nontrivial. In gravity, the r-matrix has a cubic pole and the coproduct is trivial: gravitons braid but do not split. The Drinfeld–Sokolov reduction bridges the two: it takes the affine Casimir $\Omega/((k+h^\vee)z)$ to the gravitational $(c/2)/z^3 + 2T/z$ by homological perturbation through the Sugawara nonlinearity, and in the same stroke, the ghost-number obstruction kills the nontrivial coproduct.

Datum	**Chern–Simons**	**Gravity**
boundary algebra	$V_k(\mathfrak{g})$ (affine KM)	Vir_c (Virasoro)
highest OPE pole	z^{-2} (quadratic)	z^{-4} (quartic)
A_∞ tower	strict ($m_{k\geq 3} = 0$)	infinite ($m_k \neq 0$ all k)
shadow depth	3 (class $\mathbf{L}$)	∞ (class $\mathbf{M}$)
$r^{\mathrm{coll}}(z)$	$\Omega/((k+h^\vee)z)$ (simple pole)	$(c/2)/z^3 + 2T/z$ (cubic pole)
Δ_z	nontrivial (gluons split)	primitive (gravitons braid)
κ	$\dim\mathfrak{g}\,(k + h^\vee)/(2h^\vee)$	$c/2$
$\kappa + \kappa^!$	0	13

The entire table descends from a single datum: the number of λ-powers in the λ-bracket. Affine Kac–Moody has λ^1 at most (quadratic pole), yielding strict associativity, finite shadow depth, a simple-pole r-matrix, and a nontrivial coproduct. The Virasoro algebra has λ^3 (quartic pole), yielding the full infinite A_∞ tower, infinite shadow depth, a cubic-pole r-matrix, and a primitive coproduct. The Drinfeld–Sokolov reduction is the functor that connects the two columns: it increases the λ-bracket order by two, trades the nontrivial coproduct for a ghost obstruction, and replaces the complementarity $\kappa + \kappa^! = 0$ with $\kappa + \kappa^! = 13$.

In one sentence: *gravity does not produce particles, only braids; the graviton is composite, and the ghost sector of its composite origin is the algebraic obstruction that kills the splitting.*